

After learning insertion sort and seeing that its runtime performance is $O(n^2)$, it's natural to wonder what level of improvement we can reasonably hope for. Could we find some holy grail algorithm that could sort any possible list in fewer than 10 steps? No. Every sorting algorithm will require at least n steps, because it will be necessary to consider each element of the list in turn, for each of the n elements. So any sorting algorithm will be at least $O(n)$. We cannot do better than $O(n)$, but can we do better than insertion sort's $O(n^2)$? We can. Next, we'll consider an algorithm that's known to be $O(n\log(n))$, a significant improvement over insertion sort.

Merge Sort

Merge sort is an algorithm that's much quicker than insertion sort. Just like insertion sort, merge sort contains two parts: a part that merges two lists and a part that repeatedly uses merging to accomplish the actual sorting. Let's consider the merging itself before we consider the sorting.

Suppose we have two filing cabinets that are both sorted individually but have never been compared to each other. We want to combine their contents into one final filing cabinet that is also completely sorted. We will call this task a *merge* of the two sorted filing cabinets. How should we approach this problem?

Once again, it's worthwhile to consider how we would do this with real filing cabinets before we open Python and start writing code. In this case, we can imagine having three filing cabinets in front of us: the two full, sorted filing cabinets whose files we want to merge, and a third, empty filing cabinet that we will insert files into and that will eventually contain all of the files from the original two cabinets. We can call our two original cabinets the “left” and “right” cabinets, imagining that they are placed on our left and right.

Merging

To merge, we can take the first file in both of the original cabinets simultaneously: the first left file with our left hand and the first right file with our right hand. Whichever file is lower is inserted into the new cabinet